Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with analytical characterization, spectroscopy, DOE/SPC process-development, and technical documentation experience across polymer, thin-film, and functional molecular systems. Builds reproducible process-monitoring datasets and method documentation that connect material/process parameters with quality attributes and troubleshooting decisions.

CORE COMPETENCIES

- Process Analytical Technology
- Raman/FTIR/NIR-Adjacent Spectroscopy
- Analytical Method Development
- Chromatography & MS Exposure
- DOE/SPC
- Process Monitoring & Control
- Technical Writing
- Tech Transfer Readiness

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided polymer process-development experiments, defining critical process parameters, analytical readouts, reproducibility criteria, and SOP-ready controls for scale-up decisions.
- Use FT-IR, UV-Vis-NIR-adjacent spectroscopy, thermal analysis, and SPC concepts to connect process conditions with material quality attributes and method repeatability.
- Translate process and characterization data into technical reports, troubleshooting recommendations, and documented methods for repeatable cross-functional execution.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Designed and characterized 15+ organic electrochromic material variants using UV-Vis-NIR, FT-IR, cyclic voltammetry, spectroelectrochemistry, NMR, GPC, TGA, DSC, and mass-spectrometry-informed analysis.
- Mapped structure-property-stability relationships, degradation pathways, and performance failure modes to guide molecular design and analytical interpretation.
- Built rigorous experimental documentation and datasets supporting method comparison, reproducibility assessment, and root-cause analysis.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded molecular thin-film process development, linking CVD/surface preparation with morphology, interfaces, optical performance, and reproducible process controls.
- Prepared technical updates and data packages connecting processing conditions with material performance for multidisciplinary research stakeholders.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition using DOE strategies and in-situ QCM-D process monitoring, characterizing film-growth kinetics to improve coating reproducibility.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling while coordinating multiple concurrent materials-development projects.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Process Monitoring and Analytical Readout Development
Current

Defined DOE-guided process windows, analytical readouts, reproducibility criteria, and SOP-ready specifications for scalable polymer systems.

DOE, SPC, FT-IR, UV-Vis-NIR, thermal analysis, technical reports

Spectroscopic Stability and Degradation Study
Chemistry of Materials 2025

Mapped degradation, redox-instability, and performance failure modes across functional material variants using orthogonal analytical characterization.

UV-Vis-NIR, FT-IR, CV, TGA/DSC, NMR, GPC, MS-informed analysis

Thin-Film Coating Process Development
AFOSR Funded

Connected CVD and surface-preparation variables to coating morphology, interfaces, and device-level material performance.

CVD, QCM-D, spectroscopy, process documentation

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder
2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)
2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Process Analytics: FT-IR, UV-Vis-NIR, Raman-adjacent spectroscopy, spectroscopy-based process monitoring, analytical readouts, method documentation
Analytical Methods: Chromatography/MS exposure, cyclic voltammetry, spectroelectrochemistry, TGA, DSC, NMR, GPC, QCM-D, materials characterization
Process & Validation Mindset: DOE, SPC, process windows, reproducibility studies, troubleshooting, SOPs, tech-transfer documentation, validation-package adjacency
Data & Communication: Technical reports, data visualization, cross-functional presentations, root-cause analysis, Excel, Python/basic data analysis, JMP/basic

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.

3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)